ActInf Livestream #010.0: "A variational approach to scripts" (2020)

Livestream | 1:02:46

all right
i think we're live hello everyone
welcome to the active inference live
stream
my name is daniel friedman it is
december 2nd
2020 and i am looking forward to
this discussion welcome to team com
team com is an experiment in online team
communication
learning and practice related to active
inference
you can find out about us at our website
activeinference.org
on twitter at our gmail account
our youtube channel or our public key
base team
and username this is a recorded and an
archived live stream so please provide
us with feedback so that we can improve
on our work
all backgrounds and perspectives are
welcome here and as far as video
etiquette for live streams goes
remember to mute if there's noise in
your background raise your hand so that
we can hear from everybody
and use respectful speech behavior also
just as far as schedules go
for the rest of 2020 we're going to have
four meetings
they are going to be on december 8th and
15th
discussing this paper that i'm going to
talk about right now a variational

approach to scripts
and then on december 22nd and december
29th
we will be in active stream 11
and we will read a more mathematical
paper
called sophisticated effective inference
simulating anticipatory effective
dynamics of the imagining
future so this is going to be the last
four
meetings of active stream for 2020 in
december
two on this paper two on the next paper
we have a lot planned for 2021
so if you're interested in participating
in 2021 in any way definitely
contact us if you want to and we also
look forward to doing
a few new kinds of events so if you have
an idea for an event or for a
collaboration that we could do with
activestream that'd be pretty cool
today in active stream number 10.0
finally getting up there in the double
digits
the goal will be to set the context for
the next two weeks of discussion with
everybody who wants to which is going to
be 10.1 and 10.2
the paper we're going to be discussing
is called
a variational approach to scripts by
albarazen
at all on 629 2020 on psi archive
this video as all the dot zeros are is
just an introduction or a context
for some of these ideas it isn't a

review or a meta-analysis or a final
word
and i'll really just be trying to
contextualize some of the ideas and
vocabulary of the paper
so the goal would be if you're in the
free energy principle or active
inference research community
this is going to be maybe a chance to
learn something new about scripts
certainly it was for me and then maybe
you're more familiar with scripts or
social scripts theory
at which point this hopefully would be
an inter interesting introduction
into thinking about active inference so
it is always going to be bi-directional
like that
and the punch line of this paper if i
could summarize it as such would be
active inference can unify different
conceptions of this
script concept leading to new avenues
and approaches for research
in fields such as social behavior
criminology and sexology
yep all of those are fields in
section 9.0 first we're going to go
through the keywords of the paper
then go through the aims and claims
abstract and roadmap of the paper which
are going to tell us a lot about how the
authors
lay out their argument as well as
structure what it is that they're trying
to do
in section c we will go to types and
uses of scripts

and uh draw on some quotations from the
authors
and then just look at the two figures
briefly
to understand what they might be
representing or doing in the paper
and in 10.1 and 10.2 which are the
up to up to coming weeks we will be
discussing this paper
so save and submit your questions get in
touch if you want to participate
and anyone is welcome to join first
we're going to talk about keywords
the keywords of this paper are as
follows first is active inference which
is great we always like to see that as a
keyword as well as the free energy
principle
so active inference is about the fusion
of action and inference
it brings together fields like
cybernetics ecological control theory
all these different things that we've
been talking about but maybe this is
someone's first active stream
free energy principle is a broader
physics
based or mathematics based framework
for understanding how multi-scale
systems exist
and what it is that they do active
inference is a process theory under the
free energy principle
we're not going to have slides for those
in this talk
what we are going to talk about to
contextualize the paper
philosophically and from a conceptual

perspective is
internalism and externalism as well as
the center
topic of the paper which is scripts
script theory
and specifically social scripts
we'll hear about two different
deployments or two different senses
of the script one is an internal
psychological schema
so that's like something happening in
your head maybe that would be amenable
to a psychological
or a neuropsychiatric intervention and
then another sense of scripts are in
terms of constitutive
behavioral categories so these are sort
of classifications of people
or of action sequences and they are
things that are enacted
but they may or may not hinge upon these
internal psychological schema
in fact we're going to go into a fair
amount of detail how the authors lay out
the differences amongst different uses
of the script concept but that's just to
prime it for now
let's get to this first philosophy point
but it's pretty broad it applies to a
lot of areas and that's internalism and
externalism so in active stream
5 that's 5.0 0.1 and 0.2
we specifically talked about internalism
and externalism
we talked about the paper multi-scale
integration beyond internalism and
externalism
so we talked a lot about how active

inference specifically
is able to reframe this debate or
dichotomy between
internalism and externalism or
internalism versus
externalism potentially can recast our
understanding
of how these systems work in a way
that's going to be amenable to better
understanding
control design predict etc that's what
we talked about in actin
act m5 but without going into too many
details of active inference now
let's think about three big ways that
we're going to use active inference to
transcend
this internalism and externalism debate
slash debacle and the three that we're
going to talk about right now
are one the fact that real evolutionary
adaptive systems are multi-scale
and nested so multi-scale nested systems
as a way that we can escape internalism
and externalism
a second way that we can escape
internalism and externalism
is with local causal closure and a third
route is with this uh relationship
between action and
inference so threes are three different
talking points
that you can think about are countering
the internalism versus
externalism dialectic so the first thing
is this idea that real systems
especially the ones that we care about
are multi-scale and nested they're

hierarchically embedded or they're
heteroarchical there's a lot of ways to
talk about them they're complex adaptive
systems
and the paper that we drew on in actin 5
and i'll return to here
is this paper in 2012 by dennis noble
called
a theory of biological relativity no
privileged level of causation
so his focus is on first clarifying
the so-called upward causation pathway
which is sort of a default understanding
for many people which is that smaller
things
quote cause bigger things to happen in
the sense that they compose them ergo
they must cause them
and we can look at how genes are related
to proteins and rna through a somewhat
special
connection and then how physical
entities nesting
wise compose each other into bigger and
bigger organizations
through organs organisms social
organizations
and then ecosystems technological
government schemes etc
and where noble critiques this simple
upward
causative model which lends itself to
reductionism because why wouldn't you
study the most basal
aspect of the system if that's where the
most causal potential were
uh relied upon he highlights this role
of

so-called top-down forces which are
structuring forces
from slower time scales as well as
broader scales of spatial analysis
about how states of organisms can
influence
downward in some sense smaller
things and there's the
larger versus smaller continuum that we
can still think about and to which
extent we
privilege or prioritize bottom-up or
emergent
uh phenomena versus top-down which could
also be considered emergent control
so really emergence is on both sides of
the coin but
at a given level of analysis the larger
scale can be considered external
so when the organism's epithelia is the
boundary condition
then society is external and proteins
and cellular features are internal
and if you were to go into a cell
internal would be even smaller than the
cell
almost by definition and so we can
think about how as there is no
privileged level of causation in the
sense that the
interactions within and among levels of
analysis
are what result uniquely in the system's
function and there's no way to
tune one aspect without uh necessarily
impinging upon another
we can say well yes you could
characterize things as being just

simply outside of another thing you
could say here's inside the house and
this is outside the house
so you can still have internal and
external as terms
but this idea of internalism and
externalism that the relevant causal
features are internal to an organism
versus our external to an organism
versus the interaction
it kind of is reframed in this very
fractal hierarchical framework
that the uh uh
no privilege level of causation argument
is taking us towards so that's a systems
biology approach
we're moving beyond internalism and
externalism because real world systems
are nested
a second type of argument that
evades the internalism externalism
discussion
is the idea of local causal closure
and this is figure 4 from the paper that
we discussed in
active stream 5 and it is showing an
auto catalytic
network and so here you have different
types of reactions and products
it's a metabolic network that involves
an auto catalytic
mechanism in the metabolic domain but
there's also
auto catalytic closure that might be
imagined semantically or culturally
so what this is so showing is that
because the interactions of specific
types

and the integration of ecological
resources as well
as well as multiple types of agents and
processes internal to the process
as many different features are perhaps
necessary and not alone are sufficient
to generate
the outcome of the system which is
autocatalysis you take away one of these
inputs
it's just not going to work it's like
taking away a vitamin from a human so
you take away a vitamin and they might
die
but does that mean that the vitamin is
the one and only thing that makes them
live
of course not but it does make us move
beyond this internalism
and externalism dichotomy because we
understand that all real systems are
at the interface between these two and
that's related to phrasings like
it is the nature of nurture to be
natured or
vice versa sort of just blurring the
line between them because they're such
causal dependence
for example with our microbiome the
relevance for the scripts concept here
is that
whether something like a feature is
internal or external
to one perspective or another it's part
of an integrated causal story about the
world as well as a perspective for
example
telling a story from a third part

person's perspective versus second
versus first
and another area that we might return to
is that the scripts require actors and
communities of practice to enact them
otherwise they're
not real so it's kind of like a tree
falling in a forest does anyone hear it
well
if there's a script in a file drawer is
it ever enacted
maybe that's how it enacts what it is
but
in the social sense it is not
implemented and this is uh
going to bring us to some other dramatic
drama
related dramaturgical concepts and this
idea about set and setting
and the context and the performer and
then just the last way that we can move
beyond internalism and externalism
is this fundamental relationship between
action and inference
so yeah it's playing out in nested
systems like point one
and yes uh it's also related to causal
closure in a sense like
uh point two but this one is actually
unique
in that it's saying no just
fundamentally wherever the causal
relations may be and whether this is a
one level system or a multi-level system
it's actually the fact that action and
perception are happening as an
integrated coupling
that we can debate about what is

external or what is internal
but we find that there's actually uh uh
somewhat of a reframing of this
issue of whether something that's
actually happening is outside versus
insight
okay so those are just a few internalism
and externalism questions that are more
general that are
going to be returned to when we're
talking about scripts and whether they
are internal or external to an agent
the main topic of today is this script
theory so what is a script broadly a
script is a pattern
of action and or thought and it turns
out there's a lot of ways that people
have thought about that and there's
also a lot of related topics which is
one of the reasons why
a unifying effort is being performed
here
because apparently there's a lot of
disagreement about
scripts and what they mean so terms that
i thought could be
relevant to call them besides for
scripts because they include for example
non-verbal
uh behavior something like an archetype
or a situation or scenario
a storyline or a narrative a creod
to use waddington's term from
epigenetics
so it's kind of like a pattern of action
and or thought
in one way or another depending on the
situation and the paper is going to go

on to explore
two specific axes of variation you can
think about
in the scripts concept
scripts are happening in various
contexts so first with no
citations i'll just provide a
quote from them and then a few examples
of areas where
it could be interesting so they write
the scripts construct has long history
spanning several disciplines
scripts have been applied fruitfully to
study human behavior in different fields
from disciplines centered on humans
individual human minds and their
interactions such as psychology
neuroscience and artificial intelligence
to the social sciences where has been it
has been used influentially in fields
like sociology criminology anthropology
and sexology
different definitions of scripts abound
with different focuses
the def the concept sometimes is used
under different names such as action
schemas
the idea that motivates the use of the
script construct in these scientific
approaches are
is at its core dramaturgical that means
it's related to
performance according to its proponents
what enables social agents to act in
situationally appropriate ways
is a shared set of instructions or
normative prescriptions for
situationally appropriate behavior

that's why it's called a script just
like the script guides an
actor it's sort of a loose version of a
script
that's going to not dictate exactly what
we say for example i'm not reading from
a script and most people aren't when
they're on a video call
but it's going to be a little bit
broader than that but of course not so
broad as to explain everything
the implication of this view is that in
order to act as
a cohesive social group in which every
agent knows and enacts their role
agents must share a common body of
knowledge i.e a script
that prescribes situationally
appropriate modes of being
this is metaphorically akin to actors
sharing a dramaturgical script
hence the name of the construct scripts
are used in scientific theories to shed
light on how internalized psychological
models are integrated with externalized
social models
by drawing on a pool of common styles of
performance and cognition through
contextualized acts
such as speech acts and their insured
actions driven by goals
so it's like all the world's a stage but
then they made a research field
about it we see it playing out on the
actual stage
because there's uh raised platforms
where people
perform and then there's non-raised

platforms where people perform
but there's scripts for many of these
situations sometimes literally written
out
other times more broadly
these kinds of scripts happen on the
sidewalk for example
it's interesting to think about how just
the ways that people
have uh viewed each other's bodies their
very personal space
and whether they step out of the way on
the sidewalk
depending on what someone is or isn't
wearing these are all
things that have changed over 2020 and
so we're seeing the script
update or change scripts happen in the
courtroom
whether it's the judge and the priority
that this person has by virtue of their
training and position in the room but
these types of formal
acts are places where scripts have been
applied because
in some senses the script is defined
like a religious service
and ritual spaces in general are places
where scripts can play out
and so here one can literalize
what is happening and talk about well
the angle of this object is at this
angle and the repetition of this happens
this many times
or there's an experiential element so
scripts
also reaches into this area of
experience and what makes something

important to somebody
and all that behavioral work one area
that we'll talk about in just a few
minutes
is the crime scene so scripts have been
applied to
really break down crimes in some ways
into their sub components
to understand the patterns and the
flowcharts by which certain crimes
are happening or not and also another
example might be computer scripts which
isn't really
talked about much here specifically the
entire
framework is computationalist in that
like active inference it holds us a high
value to at least be able to state
things in pseudocode or
algorithmically or as a process rather
than just an endpoint that's achieved
for example
but we can also think about these
scripts especially when they involve
our extended niche our shared
uh niche of attention and technology and
platforms
computer scripts are playing into human
scripts it's almost like one of them is
programming the other
let's return to that crime scene version
of a script
this is going to be some figures from a
book
from 2018 chapter called crime script
analysis
so this is first just a structural way
of looking at how quantitative data are

used
in the academic psychology model so we
see
behavior on the right and the arrows are
like
the hypothesized or empirically tested
correlations this is basically like a
structural equation
model but it's just saying there's all
these features
that are related for example to the
person's personality
on the top to uh things like their
behavior
and that can have many levels of detail
but this is just at the most broad level
the way that different features of the
person and the context
are statistically brought together
people developed
a little bit beyond this simple put all
their features into
a database and run a regression type
model to
this idea of a state transition diagram
which is drawing interestingly on
concepts like
markov chains and markov fields these
are things that come back
in active inference but they're used in
this
crime script community and this one
i believe shows the behaviors leading up
to drinking and driving which is a crime
and so this bottom
it says this is the drive's vehicle all
these arrows all these routes by which
they're led there

so maybe they were at home and um
then went out somewhere or they were
somewhere else and they came home
there's different funnels different flow
charts and the question
that we're going to ask here is could we
simplify and integrate this with active
inference so if we
go with no map and we just say let's
survey 500 people and then let's do some
expert analysis crowdsourcing some
natural language processing let's just
figure out
all the attributes that led to these
people uh
doing this crime then we might be able
to pull out a bunch of these nodes maybe
we could even make a flowchart like this
but could we have a more broad theory
other than just
interviewing hundreds of people always
knowing that there's going to be types
of crimes being committed that are going
to be outside of our
scope and just maybe have a more
integrated framework for
talking about how certain norms are
violated to
take into account a lot more features
and a lot more richness in the kinds of
ways that agents actually interface with
the world so that would be the goal with
scripture would be
in all these different fields that we
would be able to
simplify from just this web of
correlations
on the left and this network of flow of

all these probabilistic attributes
probably averaged out over important
subgroups
could we just take a step back and ask
about action
in general and thinking about these
specific actions
that violate formal or informal rules or
norms
could we use active inference in that
situation
here was just a social script paper that
i thought was pretty interesting
so it's called social scripts and the
three theoretical approaches to culture
by robert sinclair in louisville it says
abstract in attempting to investigate
the cultural sciences researchers have
developed
frameworks of which ratner's three
categories are most informative
the three categories are symbolic
activity theory and individualistic
approaches also social script theory is
a promising path in the integration of
symbolic and activity theory models
so that's pretty cool this is coming
from a non-active inference
community and it's a synthesis
of some of the major threads in cultural
research and again those three
categories
are the symbolic the activity theory and
the individualistic approach and
um this activity theory is kind of an
interesting thing it's
uh it's a strange idea to read about
that

people had to make a area of theory that
was about
action but it actually does touch on
things like action research today
activity theory and there's a deep
thread of this
and sometimes highlighting this thread
of activity theory is quite
useful so um here
is what is written in the paper
okay so first uh there's a discussion of
the early activity
theory and about how this activity
theory was getting integrated with the
neuroscientists the philosophers
computer scientists and cognitive
scientists
and then there's uh quoting from the
paper what is significant about the work
of these early cognitive scientists
is that they argued for a deeper
understanding of language in artificial
intelligence
shank and abelson in 1977 for example
noted that scripts protocols and
vocabularies were about language
so they naturally asked asked not only
about how computers code human behavior
but also about how languages are used to
negotiate social reality
minsky in 1985 noted that in addition to
using language
to structure frames scripts metaphors
and the society of mind
one needs to look at the mind as a
parallel processor
one capable of handling several scripts
simultaneously a situation comparable to

how human languages function within
social contexts
understandably linguistics became one of
the central disciplines within the
cognitive sciences
so that's about linguistics and computer
science and cognitive science and
artificial intelligence
and then this is a closing quote from
the paper
can the personal approach to culture
even be considered
a viable model for the cultural sciences
so that's the individualistic model
which is basically saying that the
priority of culture is just
operating individuals nest mates
it appears that the personal approach to
culture espoused by post-modernists
is not a viable model of culture who
would have supposed that
when this model is articulated by social
philosophers such as foucault badyard
and debord
the theory still lacks explanatory power
as it is not able to account for
activity theory
what constitutes activity in this model
is a passive embodiment of the signs and
symbols of mass media and mass culture
the most promising path consists in this
integration of symbolic models with
activity theory models
this is exactly what social script
theory has done so
very cool that in the social sphere the
social script theory was perceived of
as integrating the semiotics the meaning

and the communication of symbols with
the action and so
there's actually a action inference or
an action communication
action information nexus that was
integrated
in their vocabulary perhaps as social
scripts
because it has a computational bent you
know run that script
and you can even back 50 years ago
people were talking about running
multiple scripts and
the ecological dependency so these are
kind of classical ideas
that potentially under active inference
we could
uh learn more about and then just the
last
point here on this paper according to
activity theory
psychological phenomena are formed as
people engaged in socially organized
activity
since these activities are socially
formed they provide a social and
cultural influence on cognition
it should be noted that this concept of
praxis which is a german word i won't
pronounce and then
a russian word deathnost has a long
intellectual tradition
the focus on praxis in this essay is not
about that long tradition of pragmatism
and practice
but on activity theory which is
connected with vygotsky
and so it goes back to marks and angles

uh and some other
interesting directions but pretty cool
to see that there's actually the
pragmatism
and then there's an action threat so
activity research wasn't merely about
getting it done or being most effective
it was related to
that bodily concept body and motion
developmental component
by vygotsky and that led to ecological
psychology and so many other areas
so these are kind of cool threads that
are
combinations that are re-emerging within
the context of active inference
it's because what the free energy
principle and
active inference are searching for is
something to say about the relationship
between the world
and agents so here we're kind of
mutating this
common slide yet again and we're
thinking about agents who are
interacting in
a world and the world is a niche that is
shared by the agents
it's an ecological niche and that means
that it's a
niche of ecological resources as well as
shared attention
norms practice tools software
video conferencing this is where the
script concept is going to come into
play because we want something that's
computable we want something that is
ecologically responsive

adaptive can take on all these
interesting traits that we want
and yep the joint world niche is related
to their shared affordances of agents in
a niche what they can do
it's related to their umvelt what they
can perceive
as well as their culture memes norms all
these other things attention especially
and now we're going to bring this
scripts topic in
all right so the paper that we're going
to be
discussing in 10.1 and 0.2 is called a
variational
approach to scripts and the authors are
on the right side
the aims and claims of the paper are as
follows they write the aim of this paper
is to formalize the notion of script
using the modeling resources of the
active inference framework
the hope i like that it's not just a
claim there's a hope as well
is to shed light on the multifarious
uses of the construct of script as it is
used in the sciences that study
human behavior active inference is
relevant here
because it may provide the key to
formulating an integrative script
construct
so that's in their own words how they
wanted to
frame what they did in the paper and
then i would write in
non-active inference terms how can we
apply active inference to unify

the study of scripts in the social
neuro-behavioral
cultural and computational dimensions
the abstract is written here it is this
paper proposes
a formal reconstruction of the script
construct by leveraging the active
inference framework
a behavioral modeling framework that
casts action
perception emotion and attention as
processes of bayesian or variational
inference
we propose a first principles account of
the script construct that integrates its
different uses in the behavioral and
social sciences
we begin by reviewing the recent
literature that uses the script
construct
we then examine the main mathematical
and computational features
of active inference finally we leverage
the resources of active inference to
after
offer a formal model of scripts our
integrative model accounts for the dual
nature of scripts
as internal psychological schema used by
agents to make sense of event types
and as constitutive behavioral
categories that make up social order
close parenthesis and also for the
stronger and weaker
conceptions of the construct which do
and do not relate to explicit
action sequences respectively we're
going to get back to that binary

soon so the roadmap of the paper
is as follows they begin with an
introduction and talking about the
background of script
theory which was something that was a
new area
relatively for me they talked then about
internalist approaches
so remember the broader discussion of
internalism and externalism in
multi-skill systems
well here we're going to be focusing on
script theory social script theory so
the internalists
are those who are taking approaches like
psychology and neuroscience
as well as other reductionist-facing
approaches like artificial intelligence
then there'll be a section on
externalism which talks about the social
sciences which
by their nature are interested in
studying these
higher level forces larger scales of
analysis so
fields like sociology anthropology
criminology and sexology
often focus on the society looking in on
the individual
and so that is that pressure kind of
like a balloon there's pressure out
pressure in
it's actually like internalism and
externalism and there's a membrane
could it be a markov blanket who knows
we'll then uh move to the other
axis beyond internalism and externalism
is a strong versus weak conception of

scripts which we'll go through
they talk about the two extreme
perspectives
then they talk about variations on
scripts which
are kind of what you would expect the
performance
components that relate to spontaneity
and ecology
and mutational variations all these
types of things that cause scripts to
vary
after introducing script theory they
move to the abcs of active inference
by providing first a introduction to
active inference which is
perhaps great for us all to check out
and then
talk about how shared generative models
can be
thought of and how sociocultural
dynamics can be thought about
in the culture-generating way as shared
models
figure one then has a simple generative
model for policy selection
which is a figure that we'll get to and
then figure two
is a heuristic description of the
generative model of the niche and of the
agent
section four as is where they come
together
and talk about an active inference
account of scripts
so as usual introduction area a
then active inference then we bring
those two areas together

in section four or five or so and in the
active inference account of scripts
there's a discussion of scripts as
shared conceptual niches about types of
events
as well as scripts as representations of
typified sequences of events
so blurring the idea of types and events
in a way that we'll get to and then
there's a
discussion and a conclusion cool so
that's the
roadmap i think that in this
contextualizing video
what would be helpful is to characterize
not just the breadth of where scripts
can apply or have applied or were useful
weren't useful
but zoom in on these two dimensions of
variabilities and the specific problems
that they cause in that field
and then how what active inference does
will end up alleviating perhaps some of
these tensions
this is about scripts and their
discontents
so here's what they write there are some
issues that stand in the way of an
integrated
model of scripts across fields the first
is that the concept gets implemented
differently
in different theories and disciplines
which throws down on our ability to
provide a unique definition that can do
justice to all the use cases of the term
in the literature
similarly different terms can be used to

describe very similar phenomena across
domains of study
like script or schema prepended with
terms with
with terms like socio cultural or
cognitive
some authors have attempted to unify the
concept in our view these attempts have
had rather limited success
as the varied census of the term suggest
so what's the issue
the issue is that scripps doesn't have a
coherence
implementation and it also doesn't have
a coherent definition
now whether there is one chicken in the
egg or egg in the chicken
that's another question and
remembering that what we want to do is
explain
predict design control scripts just like
any other system
we would want to have the best and the
most flexible understanding
of scripts that would capture all of the
diverse situations where scripts are
being deployed in
but also integrated with more general
theories
so maybe sociophysics maybe some other
quantitative
frameworks and they're going to
focus on two primary dimensions
of variation i'll read the quote
first in this paper we focus on two
orthogonal distinctions that we suggest
structure discussion on scripts in the
literature the first is a split between

internalist and externalist readings of
the construct of the script
on the internalist conception a script
is defined as a cognitive structure that
is typically internal to an agent eg
encoded in their brain and that
harnesses information about typified
behavioral patterns that are appropriate
in specific social situations
on the other externalist conception
scripts are cast as the basic fabric
from which social institutions are
crafted
on this conception a script is a set of
highly codified
practices and rules
practices norms standards beliefs and
linguistic practices
and rules that make up an institution
some conceptions are not easy to split
between internalist and externalist but
the way in which an agent interacts and
reproduces a script does seem to bear
elements of the duality nonetheless
and so that's where we return to these
two extreme cases that were presented
earlier
which is the ultimate internalist versus
externalist the internal
psychological schema is the internalist
perspective
and it ultimately is going to tend
towards neural
reductionist perspectives the
constitutive behavioral categories
perspective
is the externalist perspective bigger
system society influencing the

individual
and that tends towards holism these are
the two
tensions to poles of the script theory
and just as we've seen before
we want to use active inference to
reframe this variation
along this axis from systems where it
really does just plainly appear that
it's a small thing that's bubbling up to
cause a larger change
or vice versa we want to take a new
perspective on this question about the
localization of scripts
so that we can be better and act better
in the world
and the second dimension is a split
between the weak and the strong
readings of this term and the readings
differ
on how to explicitly uh think about the
way that a script prescribes
specific courses of action for the
agents
so in the strong concept of the script
a script is a list of explicit
instructions for situationally
appropriate behavior
either neurally encoded under the
internalist reading
or implicit in conventions maintained by
the institution
in the externalist reading the strong
reading
of scripts dovetails with work in motor
control
that casts the process of motor control
as the execution of a motor

representation
which is cast as a list of explicit
instructions for action
so kind of like robotics the weak
reading of the script construct relaxes
the assumption that a script
prescribes the precise order of events
that it entails
a weak script just encodes or harnesses
information about the kinds of factors
that an agent might encounter in a given
situation type so let's graphically
lay out these different variants of the
script theory
so on the x-axis is from weak to strong
so weak is
it's a tendency it's kind of related to
information little
quantitative amounts of reduction of
uncertainty or preferences
that are guiding bias implicit bias
whatever it happens to be
and then on the strong side it has to do
with the certainty which a certain
action is carried out
so maybe some events are over on that
end of the spectrum like a
dramatic performance and then the y-axis
is this internalist to externalist
debate or tension where internalist
things are smaller
more inside of the epithelium more
reductionist perspective
externalist tends to be outside of the
epithelium bigger things
more holist so a strong externalist
perspective would be like a marionette
show or a puppet show because the

behavior of these individuals
is going to be driven by external forces
to them
and it's going to be very structured and
deterministic
in some way it could even be done by a
machine
holding the wires for example whereas a
weak
externalist situation might be something
like people who are in a meeting and the
body language or the ways in which
certain people
fit into socially prescribed or
different cultural perceptions on a
certain
presentation that someone provides all
of these
different soft features could end up
playing
into the behavior of individuals that
would be a very externally
oriented thing but also there's the
collective behavior in the ecology of
the group
so that's still externalist but it ends
up being more probabilistic
the strong internalist beliefs relate
to highly computational ways of thinking
about the brain
and about the ways that certain action
patterns are encoded
based upon representation as was stated
in the
earlier quote and then the weak
internalist perspective
might see the mind as a
chaotic dynamical system or something

that's never returning to
steady states or has a strange loop
feature
all these different aspects that make
even though
it is a focus on the psychology um
so it could lean towards reductionism
there's also
a broader understanding about the
potentially emerging components of even
internal features
all these different ways that we can
think about scripts and then
here's fristen the origin
here and going to integrate across these
quadrants
with the active inference framework
let's just look at one of the quotes in
the paper
they write our proposed formalization of
the script construct
via active inference allows for
interesting avenues in social computing
specifically we can begin to make
predictions about how humans react to
scripts
by clearly identifying the formal role
of internal and external script elements
as well as what weak scripts and strong
scripts entail in a cognitive and
ecological structure
we can begin to leverage the model to
identify the moment-to-moment dynamics
of interactions
between social agents in a given context
we can identify how narratives
influence expected behavioral and
contextual framing

cool with our formal model of scripts
we can map the direct interaction of an
individual with social categories and
events
as well as its con commitments in the
shared niche
those dynamic
those dynamics could not only
could allow not only to model how
scripts come to be widespread
by simulating sensitivity to deontic
cues and social coordination with
interlocutors
but also how scripts may come to change
when an agent is faced with a script
violation
and its different ramifications these
violations may be met with social
punishment
or be embraced when they tap into the
previously invisible
valuable social reality i'd love to hear
more from the authors about that
part this formalism may be scaled up to
simulate agents as the niche and see how
certain patterns of interaction and
co-option may
emerge also really cool idea finally
the model and the predicted patterns may
be measured against real empirical data
and falsified or confirmed to test
psychological hypotheses
about adherence to scripts
so all the areas that we care about
especially in the social areas
anywhere you can cast broadly a strong
versus weak or an internalism versus
externalism

something you're basically talking about
the spatial and the temporal scales and
the systems that humans care about
so if you're here and you're from script
theory and this is active inference
then maybe ask questions about
what would make it useful for your
communities or
for the specific situations uh the
script construct
is something that also has to be enacted
by a mental health worker or
by a social worker by a government or by
a startup so
what tools exist from the script
community to help us structure those
kinds of
flows so let's try to think about just
bridging between our different
communities and
about these different approaches to
research and
learning from each other because i know
that the scripps community probably has
so much to
share about how these tools
and models that they've used have worked
how has not worked
how can we use active inference to
confirm what they know
and then also have a understanding about
why things that they thought would work
but didn't work
didn't work or why things they didn't
expect to work
did work can we frame it in a way that's
based upon their surprise in the world
as actors with deep generative models

and then also zoom out and think about
the social context as
being something that responds perhaps
non-linearly to interventions
so these would be kind of fun things to
talk about
we've had a lot of social sciences
people on the active stream before so
if that's something that anyone knows
about script theory
or any of these different areas of
anthropology
then it'd be awesome to hear from them
because
like i said a lot that we could learn as
an active inference community about how
these theories
have taken hold in different areas and
come to
make impact in people's life so this is
a real
interesting bridge between people's
lived experience
and the mental health uses of scripts
the crime
dynamics and the way that governance is
tied in with it so
really interesting stuff um i hope that
people
can read the paper and attend one of the
upcoming discussions or if they
are seeing this video after we had the
discussion then just leave your comments
because
you know the discussion just doesn't
stop so
just to go through the figures um
this is figure one and i added colors

to clarify a few of the different things
that are happening here
so the figure's caption reads as follows
it reads as
a simple generative model for policy
selection this schematic
depicts a generative model for policy
selection it represents probabilistic
beliefs about
how observations are related to the
states that cause them
that's the likelihood matrix a
there's also in this problem in this
probabilistic model there are beliefs
about the manner
in which states of the world evolve over
time which is the state transition
matrix b
and then there's beliefs about states
prior to sampling the world
which are green and they're indeed and
then preference
over outcomes c is not depicted here but
as written on the top
the calculation of g is related to the
calculation of c the outcome preference
matrix
and also things like affordances what
the agent can do
and then this g is what modifies policy
selection
so perhaps someone coming from the
social sciences
wouldn't have expected this kind of a
technical diagram to be
in a paper about scripts so let's just
think about what everything is doing in
this figure

and what all the edges and nodes
represent because we've talked about
this figure in more technical
active streams for example like number
eight or like the upcoming number 11
but here let's think about what this
does for script theory
so each of these letters are like a kind
of node so the two blue nodes are
similar
the kinds of red nodes are similar and
the d is an initial condition node
so everything that has a similar shape
or coloration is like one type of thing
and this figure reads from left to right
from the
initial conditions that's like time
equals zero
through different time steps and at each
time step
there are several things that are
happening one thing that's not shown
here many things are not shown here this
is a pretty reduced format
of the formalism but what's happening is
that the agent
is maintaining a state estimate through
time
and so that could be like uh what kind
of relationship
am i in or what kind of social situation
am i in
is this a friendly conversation or is
this a fight that could be a zero or a
one
or the state estimate could be a lot
more high dimensional a lot more nuanced
and the state estimate is being carried

out
through time well what else is tied to
the state estimate at each moment
one thing it's tied to is oh
observations so that could be any kind
of observation it could be the person's
tone of voice or their facial expression
or the content of their speech
or it could be an observation in a text
message that says
hey this person is not honest or
something like that
so many kinds of observations again this
is just a big
broad sketch but at each time step
the state estimate which is am i in this
type of conversation or that type of
conversation
again whatever it happens to be the
state estimate
is being mapped to the observation
we're not going to worry about the
mathematical details now
but it turns out that what connects the
state estimate
to the observation is a belief
about how the observations are related
to the states that cause them
so there's some sort of internal mapping
let's just say again sidestepping this
debate that we had a little bit earlier
that um smiles mean the person is happy
um and frowns mean they don't but then
you think a little deeper and you
realize well it's uh some if someone's
trying to deceive you then they might
smile
so that might trigger your subconscious

receptors your subconscious priors but
maybe you look beyond those types of uh
physiological responses so again there's
layers of nuance that aren't going to be
in this
figure but just broadly the state
estimate is
updating through time as a function
mapped to the observations that you're
expecting
and estimating to get the blue
b is the beliefs about the manner in
which states of the world evolve over
time so here's like
the person's happy they're smiling my
policy my action selection is that i
if i say something nice then i estimate
that they'll continue to be happy and
then they'll continue to emit
this like happy sound and then here if i
continue this policy so this is at
zero time zero one two three so and this
is like the first three moves of chess
so you get starting state and then
observations first round
a movement guided by a policy
this is also framed as a belief about
how the world is expected to change
so under certain estimates about how the
world is
how will my trades today in
cryptocurrency
manifest as future observations or
given everything i know about the person
all these norms and the social
components
how will my uh policy on a certain type
of conversation

end up changing future observations i
have and then it turns out that in the
free energy
principle slash active inference
framework the policy
that's the whole control theory
cybernetics elements to the theory
that is guided by g which is
framed a few different ways and it's
kind of situational
how exactly it's deployed but there's a
lot of
mathematical work in other live streams
and in
other talks other papers to go into this
what is g
and what do all of its calculations mean
but basically it's a heuristic that
helps the organism
select good policy and good policy is
not
absolute because there's scripts for
social failure
there's scripts for success and so
instead of getting into this realm of
well what if the person
consciously wants to succeed and all
this sort of stuff
we're going to take a statistical
approach and we're going to have this g
formalism that turns out to map onto the
way
that policy is selected even if it's in
a distorted way or even a pathological
way
okay here is a quote from the authors
they write
with this in place it becomes possible

to implement weak scripts in a
generative model
we submit that weak scripts can be
implemented via the likelihood mappings
a
prior beliefs d and sensory preferences
c of the agent
thus weak scripts harness beliefs about
how
the expected salient social categories
for figures in specific situations
that's d
and beliefs about how they generate
sensory behavior a
so when people are feeling this way
socially then they respond this way from
an observations
way this is what you'll see you can
imagine where that becomes pathological
or whether there's challenges with
intercultural or intergenerational
communication
in social situations the relevant social
categories of role
and appropriate behavior can only be
inferred which requires the agent to
mobilize the right kind of knowledge
so it's not just enough to know you
can't just know the 1001 phrases there's
also an element of
optimal experimentation in the space and
being an improvisational partner you
can't just go off of a
hard-coded script an agent must infer
the proper categories the proper
associations and the proper mapping onto
observations in order to navigate a
social context adequately

and to maximize her returns by the niche
social capital this mapping changes
in function of the context changes as a
function of the context
hence the weak script also feeds one
understanding of the context per se
so that's many ideas we've talked about
with the environment influencing the
individual and the individual
influencing the environment often in a
way that changes the probability of
other things happening later
that stigma g this also brings up this
multi-scale idea
where within an agent statistical
uh changes can be happening and those
can be resulting in changes in behavior
all these different things so i just
thought might be interesting because
these kinds of figures are not usually
found in the scripts
papers that i looked through one other
figure that will just be kind of quickly
go through not reading the
caption is just to look at this agent in
the world
model and just how different types of
things are categorized on here so on the
bottom of the image
laser on the bottom of the image is the
world
and then the top of the image this box
is the person and so
perhaps there's some type of blanket or
some mediating interface
happening here and there's several types
of things
there are strong and weak components so

using the previous dichotomy of some
things that are sort of
objectively a specific way and then
other things are more probabilistic
in nature so perhaps this is reflecting
the observations
coming in with a strong component here
maybe that's just your
visual reception it could be debated
whether that's a strong component or not
and then it's get getting passed off to
these weaker components maybe
conditional ones or ones that are
related to stochastic processes and then
there's a mapping from action back onto
the world and the causal process of the
world
which might be a strong component or a
weak component and the point is
uh it shouldn't even have to be a binary
per se
it should be about our inference of
those latent links in the world
maybe for a certain thing you have
enough information or you have the right
side information
or you have the right prediction in a
given scenario where you can confirm it
post hoc where you can perform optimally
as if you knew
the actual strong causes of an event
other times it might be possible to
find a pareto front with respect to
policy where even if there's a long tail
distribution
of uh randomly distributed events it can
still be possible to have
incredible performance in real world

settings for example
companies that are high reliability
organizations errors do happen they just
don't
cascade in these organizations so how
could we construct
niche where different parts of this
system
and the actual unique interfaces that
are numbered here
and described a little bit more in the
caption
how could we design do systems
engineering in this kind of an
environment
and potentially do things like reduce
error rates or
increase the accessibility or onboarding
onto research communities
you know just the kinds of things that
we might want to care about
all the social scripts all the reasons
why people have been talking in pages
and pages and pages of
social sciences talking about different
categories well
here might be the opportunity to
actually specify these categories
we could debate whether specifying them
is a good or a bad thing
i would say if we all work together on
it then it's a good thing to formalize
it
maybe there's people who disagree so for
those who don't want to formalize script
theory
i'm not sure would be interesting to
talk to them here

but we're going to be talking about how
active inference
could lead to some type of integration
here and then what that might allow
so let's just think about a few general
questions in closing here
on the hour first what might
a framework for scripts be so
two axes of variations were laid out but
there's other axes of variation and it
was acknowledged that
even sometimes it was hard to
differentiate a theory as being
somewhere on the internalist externalist
spectrum
and so what what will a framework for
scripts
be how will we know it when we see it
will it look like an equation will look
like four equations that can't be
reduced will it look like 11 will we
believe that later on it will be reduced
or will we believe that later on it
won't be reduced
so what are we really going for here
with this scripps theory
what might we be able to do with a good
script
framework for example in online teams or
in uh located situations could we
do design in a way that was transparent
for
online behaviors so that people knew
that they were signing up for a service
that was
using such and such uh in its decision
making for its algorithm
would people want to be interfaced that

way
in api for a person as one of my friends
puts it
could it be possible for humans and
computers to co-program
how formal should the scripps concept be
if it turns out to explain a tremendous
amount of behavioral variation
then it might be possible to design
another
level of control for people it's kind of
like surveillance capitalism did a lot
with just linear models and
did a lot with machine learning and is
the next level just the next level of
that
is that really what we're working
towards i don't think so
one might want to ask what are the
unique predictions and experiments
so given that you think about scripts
now from an active inference perspective
what are the ways in which we can look
at the previous
literature on scripts reinterpret it and
then think
well knowing what they knew and the way
that they phrased it and
building on it and then knowing what we
know from other areas
what would be the most informative
experiment to run
so what predictions would we have that
haven't been analyzed yet but already we
can do it with the data we have
like we could predict that there's uh
more conformity in certain types of
interactions than others

and we could have an a priori way of
pre-registering a hypothesis and then
doing a deep dive into a social data set
coming up with these answers that would
be not predicted by someone else's
scripts theory or impossible to predict
with someone else's scripts theory
and showing that it could predict well
with active inference but also then
there's the
experimental design and so that's about
what
um experiment is going to optimally
reduce
your uncertainty about something
specific so it's always going to be
experiments
not just open-ended but about learning
something specific
and so i would be curious just in the
active inference community or
the social side what are people curious
to reduce their uncertainty about
what do we want to test this on do we
want to have this
tested um in which field
can it be done with data that already
exist or could exist
which is related to the question what
are the next steps for active inference
here
so there's two figures and there's uh
maybe an appendix in the paper and
uh it lays out qualitatively
mostly an agenda for formalizing and
integrating the scripts concept
and so i just wondered what are the next
steps is it actually to

pick a canonical type of interaction
ordering food or something like that
and uh chat bought it or do something
with robots or
is this gonna be something that we can
infer
from video data is it going to be on
online teams or
located is it going to be more
mathematical how can we make active
inference better account
for not just sort of robot foraging
behavior but
human behavior and we've talked about in
the last couple of weeks
people who are implementing things like
daydreaming access
and memory and historiosity and risk
averse
optimization how can we start bringing
some of these topics into play
and then thinking about real social
issues
and how we might be using active
inference to
act in those situations and then lastly
just as always we want to consider
the goals of this research and uh
i'd be curious to hear what the author's
goals or interests are uh i think it's
kind of a
cool area they're making
interesting progress sounds really cool
to integrate across
big areas of research and
that's it kind of an interesting paper
to read
so i hope everyone can check it out and

join a discussion
thanks for participating there will be
follow-up forms for the live
participants
and any feedback suggestions questions
always welcome
stay in communication with us but other
than that
thanks for listening to 10.0
you're always welcome to come on the
live discussion
or plan some other type of event and
leave us a comment or let us know if you
have any questions anything we could
address
in another show so
have a good day everyone and i'll talk
to you later